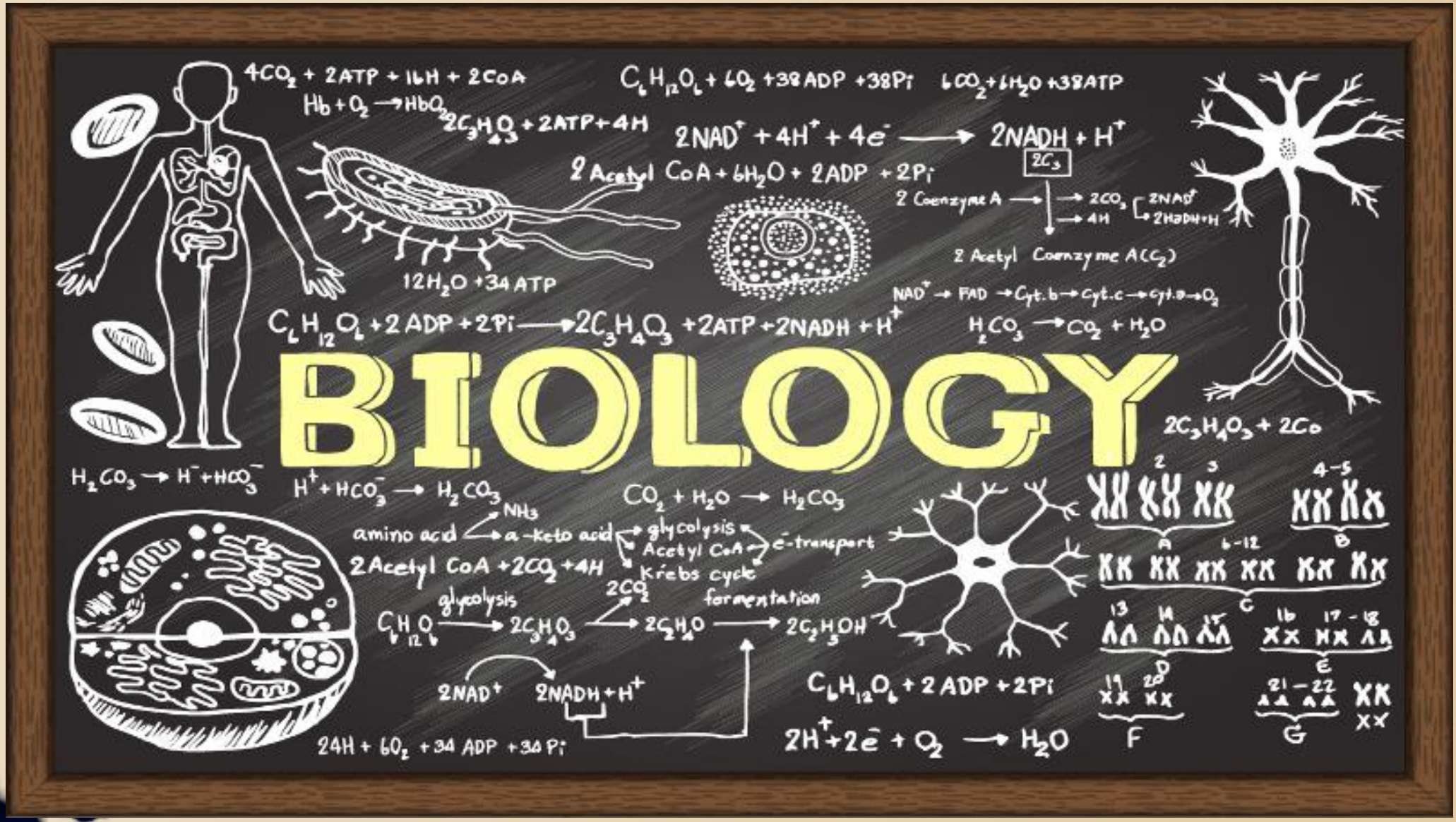
1800 – 1900


BIOLOGY
4CO2 + 2ATP + 16H + 2CoA
Hb + O2 → HbO2
2C3H4O3 + 2ATP + 4H
C6H12O6 + 6O2 + 38ADP + 38Pi
6CO2 + 6H2O + 38ATP
2NAD+ + 4H+ + 4e- → 2NADH + H+
2 Acetyl CoA + 6H2O + 2ADP + 2Pi
2 Coenzyme A
2CO2
4H
2NAD+
2NADH+H
2 Acetyl Coenzyme A(C2)
12H2O + 34 ATP
NAD+ → FAD → Cyt.b → Cyt.c → Cyt.a → O2
C6H12O6 + 2ADP + 2Pi → 2C3H4O3 + 2ATP + 2NADH + H+
H2CO3 → CO2 + H2O
2C3H4O3 + 2Co
H2CO3 → H- + HCO3-
H+ + HCO3- → H2CO3
CO2 + H2O → H2CO3
NH3
amino acid
a-keto acid
glycolysis
Acetyl CoA
e-transport
Krebs cycle
2Acetyl CoA + 2CO2 + 4H
glycolysis
2CO2
fermentation
C6H12O6 → 2C3H4O3 → 2C2H4O → 2C2H5OH
2NAD+
2NADH + H+
C6H12O6 + 2ADP + 2Pi
24H + 6O2 + 34 ADP + 34 Pi
2H+ + 2e- + O2 → H2O
A
B
C
D
E
F
G

Outline

What we know as Biology (life study) today was known as Natural Philosophy until well into the 17 th century.

Scholars of Natural Philosophy concentrated on exploration, discovery, and taxonomy (the classification of the species)

Like pre-modern physics and chemistry, progress in Natural Philosophy was limited to that which could be seen by the eye and manipulated by the hand.

In the 19 th Century, Darwinism focused on variation, struggle for existence, and natural selection

Key to the emergence of modern Biology in the 19th century was the development and refinement of the light microscope.

Light Microscopes

With continual improvements made to microscopes over time, magnification technology advanced enough to discover cells in the 17th century.

This discovery is largely attributed to Robert Hooke, and began the scientific study of cells, also known as cell biology.

Over a century later, many debates about cells began amongst scientists.

Nehemiah Grew published several volumes on the Anatomy of Plants (1682) exploring various parts of plants with the microscope.

Progress of Microscopy

- All these scientists was limited by serious problems of the 17 th and 18 th century microscopes.
  - Poor quality glass
  - Lens acting as prism that distorts light (chromatic effect)
- Improved lenses done by Anton van Leeuwenhoek and later the invention of compound lenses magnified objects 500 times
- Later the electron microscope marvelously advanced biological research.
  - 20 times as sensitive as the light microscope...

Cell Theory

The idea of cells as a fundamental unit of life was eventually formulated in 1839.

This is usually credited to Matthias Schleiden and Theodor Schwann. However, many other scientists like Rudolf Virchow contributed to the theory.

Cell theory has become the foundation of biology and is the most widely accepted explanation of the function of cells

Plant Cell

Schleiden (1860s) believed biology must deal with the actual structures of plants and animals examined

Schleiden concluded that the plant cell was the key to understanding botany, and that the **nucleus** was the universal elementary organ of vegetables

He concluded that all aspects of plant physiology were fundamentally manifestations of the vital activity of cells

Animal Cells

Schwann studied animals. However, animal cells were quite transparent therefore it was harder to study

Like Schleiden, Schwann found the **nucleus** is the key to understanding the cellular structure of animals

Schwann had noted some similarity between plant and animal cells, but he was most impressed by their differences and by the great variety of animal cells compared with plant cells

Unification of Animal and Plant Cells

With Schleiden's help, however, Schwann understood that

- There was a certain unity in plant and animal cells
- Cells are the basis of all animal tissue, no matter how specialized
- All living organisms are composed of one or more cells
- The cell is the basic unit of structure and organization in organism

Cell Theory

Rudolf Virchow studied medicine and focused on the cellular propagation and concluded that in normal growth, cells propagate from the division of parent cells.

Furthermore, diseased cells also grow or multiply from pre-existing cells

Extensive studies of cancer convinced Virchow that cancer cells differ from normal cells primarily in behavior rather than in structure.

Cell Division

By 1870, scientists identified the first steps in cell division in the nucleus (mitosis)

By the end of the century, using new staining techniques, chromosomes were discovered and their key function as agents of reproduction were identified.

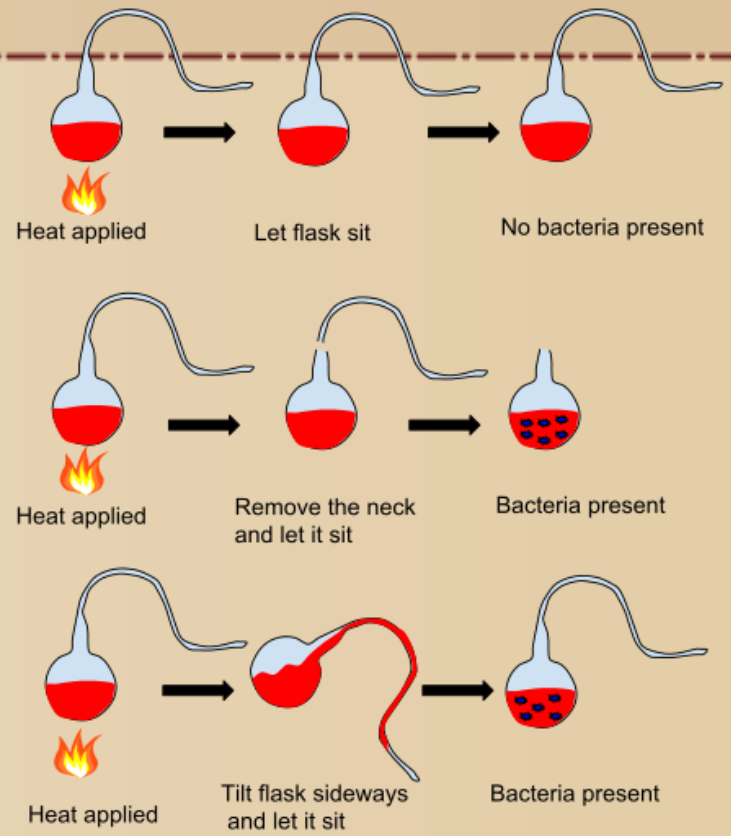
Miasma Theory

Disease transmission held that diseases such as cholera, chlamydia or the Black Death were caused by a miasma (μίασμα, Ancient Greek: "pollution"), a noxious form of "bad air". Such infection was not passed between individuals but would affect individuals within the locale that gave rise to such vapors.

Transitional period began in the late 1850s as the work of Louis Pasteur and Robert Koch resulted in the discovery of bacteria

Louis Pasteur – Germ Theory

Louis Pasteur's pasteurization experiment illustrates the fact that the spoilage of liquid was caused by particles in the air rather than the air itself. These experiments were important pieces of evidence supporting the idea of Germ Theory of Disease.



In the 1870s, Joseph Lister was instrumental in developing practical applications of the germ theory of disease with respect to sanitation in medical settings and aseptic surgical technique.

Partly using carbolic acid (phenol) as an antiseptic.

Charles Darwin 1809-

Charles Darwin was born in England from a wealthy family.

He went to Edinburgh to study Medicine, but he did not like it and could not get acceptance.

He studied religion at Christ College but did not do well with Greek and Latin.

He studied Geology with the recommendation of Professor Henslow of Cambridge and became a botanist.

Expeditions

He was invited to join the expedition to survey and observe the coast of South America. It was common to carry a naturalist on board for such expeditions. Darwin's duties were to observe, note, collect specimens and data on geology, botany, and zoology.

He seems to have been most enthralled by the giant tortoises and sea iguana he saw in the Galapagos

Darwin published 25 books. The most notable ones:

- The Structure and Distribution of Coral Reefs, 1842;
- The Voyage of the Beagle, in several parts 1839–1845;
- On the Origin of Species by Means of Natural Selection, 1859;

Struggle for Survival

Darwin noted on the struggle for survival.

Struggle for existence inevitably increase variety

Every being must suffer destruction during some period of its life. Otherwise, on the principal of geometrical increase, no land could support the product

Natural Selection

Darwin's mechanism of natural selection is based upon three principles:

1. Populations grow exponentially
2. Variation occurs within species
3. Traits can be inherited

Given these three principles, over a vast number of generations there is a natural selection of those species which have the best survival characteristics.

Slowly through this process the species evolves to be better suited to the environment. Eventually it has become another species.

Descent with Modification

Darwin's theory of Descent with Modification was developed without knowledge of genetics or the laws of inheritance.

Gregor Mendel published the results of his genetic studies in 1866

Mendel's work remained virtually unnoticed until 1900.

The first half century of debate regarding natural selection and evolution took place without any understanding of genetics.

Watch the Documentary

Origin of Life - How Life Started on Earth

https://www.youtube.com/watch?v=xyhZcEY5PCQ